Universidad Central del Ecuador

Facultad de Filosofía Letras y Ciencias de la Educación

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Curso: 5to "B"

Fecha:

Vectores (20)

1) $\vec{A} = 3i + 4j$ (m)
$\vec{B} = 5i + 2j$ (m)
$\vec{C} = 2i + 6j$ (m)

$A_x = 3$, $A_y = 4$ (m)
$B_x = 5$, $B_y = 2$ (m)
$C_x = 2$, $C_y = 6$ (m)

$\vec{R} = \vec{A} + \vec{B} + \vec{C}$

$R_x = A_x + B_x + C_x$
$R_x = 3 + 5 + 2$
$R_x = 10$ (m)

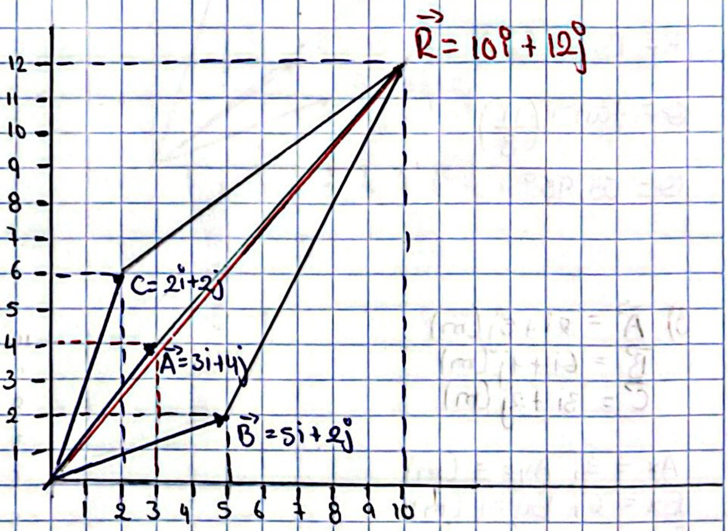
$R_y = A_y + B_y + C_y$
$R_y = 4 + 2 + 6$
$R_y = 12$ (m)
$\vec{R} = 10i + 12j$ (m)

$R = \sqrt{R_x^2 + R_y^2}$
$R = \sqrt{10^2 + 12^2}$
$R = \sqrt{244}$
$R = \sqrt{15,62}$

$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right)$

$\theta = \tan^{-1}\left(\frac{12}{10}\right)$

$\theta = 50.19°$



2) $\vec{A} = 4i + 1j$ (m)
$\vec{B} = 1i + 7j$ (m)
$\vec{C} = 3i + 3j$ (m)

$A_x = 4$, $A_y = 1$ (m)
$B_x = 1$, $B_y = 7$ (m)
$C_x = 3$, $C_y = 3$ (m)

$\vec{R} = \vec{A} + \vec{B} + \vec{C}$

$R_x = 4 + 1 + 3$
$R_x = 8$ (m)

$R_y = 1 + 7 + 3$
$R_y = 11$ (m)

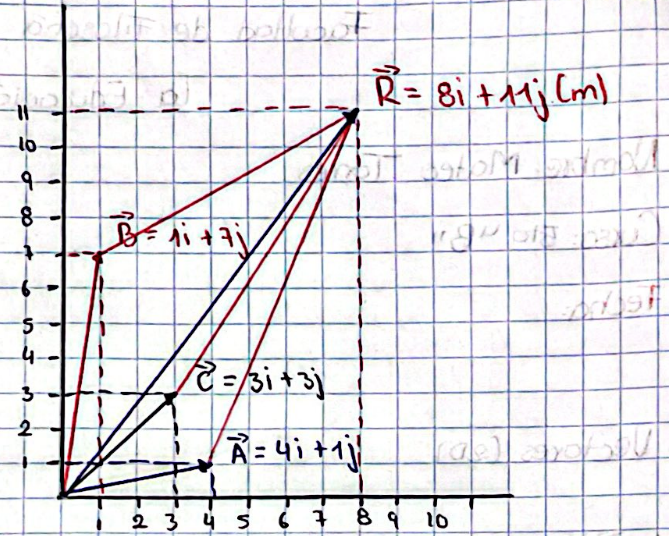
$\vec{R} = 8i + 11j$ (m)

$R = \sqrt{8^2 + 11^2}$
$R = \sqrt{64 + 121}$
$R = \sqrt{185}$

$R = 13.60$ (m)

$\theta = \tan^{-1}\left(\frac{11}{8}\right)$

$\theta = 53{,}95°$



3) $\vec{A} = 2i + 5j$ (m)
$\vec{B} = 6i + 1j$ (m)
$\vec{C} = 3i + 2j$ (m)

$A_x = 2$, $A_y = 5$ (m)
$B_x = 6$, $B_y = 1$ (m)
$C_x = 3$, $C_y = 2$ (m)

$\vec{R} = \vec{A} + \vec{B} + \vec{C}$

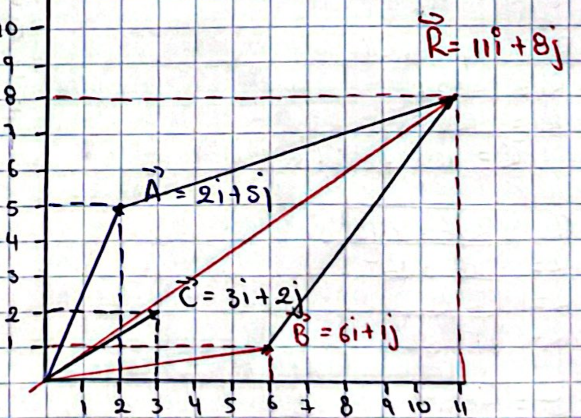
$R_x = 2 + 6 + 3$
$R_x = 11$ (m)

$R_y = 5 + 1 + 2$
$R_y = 8$ (m)

$\vec{R} = 11i + 8j$ (m)

$R = \sqrt{11^2 + 8^2}$

$R = \sqrt{121 + 64}$



$R = \sqrt{125}$

$R = 13{,}60\ (m)$

$\theta = \tan^{-1}\left(\frac{8}{11}\right)$

$\theta = 36{,}03°$

4) $\vec{A} = 5i + 2j\ (m)$
$\vec{B} = 2i + 6j\ (m)$
$\vec{C} = 4i + 3j\ (m)$

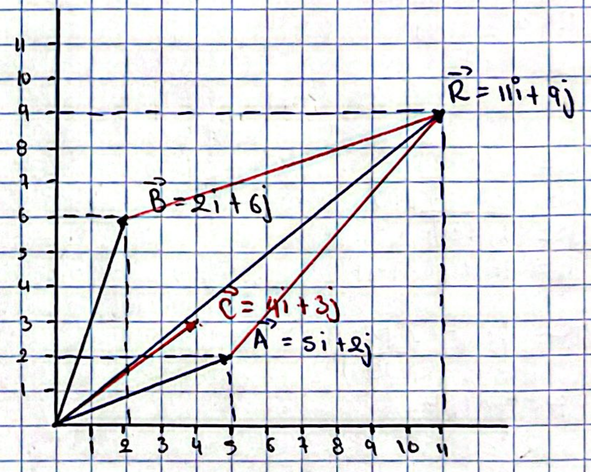
$A_x = 5,\ A_y = 2\ (m)$
$B_x = 2,\ B_y = 6\ (m)$
$C_x = 4,\ C_y = 1\ (m)$

$\vec{R} = \vec{A} + \vec{B} + \vec{C}$

$R_x = 5 + 2 + 4$
$R_x = 11\ (m)$

$R_y = 2 + 6 + 1$
$R_y = 9\ (m)$

$\vec{R} = 11i + 9j\ (m)$



$R = \sqrt{11^2 + 9^2}$

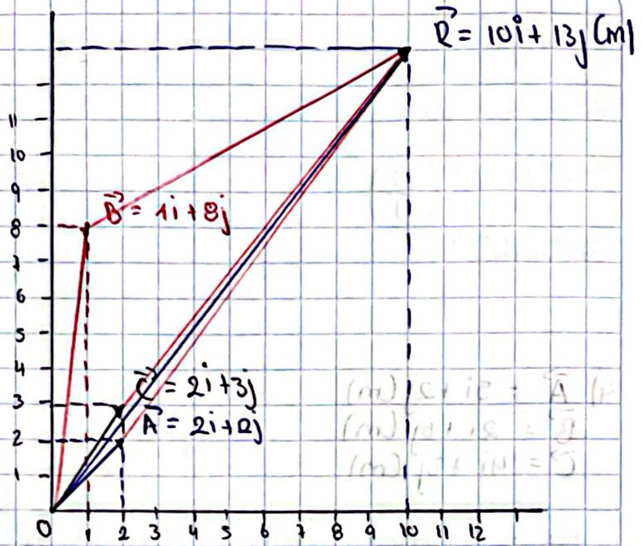
$R = \sqrt{121 + 81}$

$R = \sqrt{202}$
$R = 14{,}21\ (m)$

$\theta = \tan^{-1}\left(\frac{9}{11}\right)$

$\theta = 39.29°$

5) $\vec{A} = 2\hat{i} + 2\hat{j}$ (m)
$\vec{B} = 1\hat{i} + 8\hat{j}$ (m)
$\vec{C} = 2\hat{i} + 3\hat{j}$ (m)



$A_x = 7,\ A_y = 2$ (m)
$B_x = 1,\ B_y = 8$ (m)
$C_x = 2,\ C_y = 3$ (m)

$\vec{R} = \vec{A} + \vec{B} + \vec{C}$

$R_x = 7 + 1 + 2$
$R_x = 10$ (m)

$R_y = 2 + 8 + 3$
$R_y = 13$ (m)

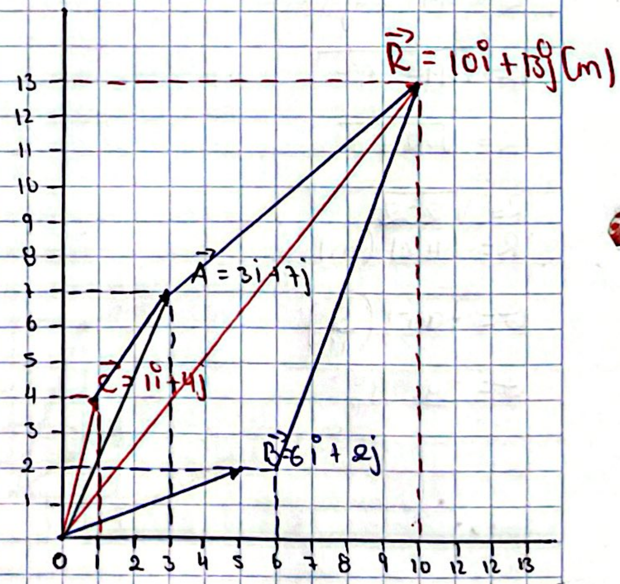
$\vec{R} = 10\hat{i} + 13\hat{j}$ (m)

$R = \sqrt{10^2 + 13^2}$
$R = \sqrt{169 + 100}$
$R = \sqrt{269}$
$R = 16{,}40$ (m)

$\theta = \tan^{-1}\left(\frac{13}{10}\right)$

$\theta = 52{,}43^\circ$

6) $\vec{A} = 3\hat{i} + 7\hat{j}$ (m)
$\vec{B} = 6\hat{i} + 2\hat{j}$ (m)
$\vec{C} = 1\hat{i} + 4\hat{j}$ (m)



$A_x = 3\quad A_y = 7$ (m)
$B_x = 6,\ B_y = 2$ (m)
$C_x = 1,\ C_y = 4$ (m)

$\vec{R} = \vec{A} + \vec{B} + \vec{C}$

$R_x = 3 + 6 + 1$
$R_x = 10$ (m)

$\alpha = \tan^{-1}\left(\frac{13}{10}\right)$

$\theta = 52{,}43^\circ$

$R_y = 7 + 2 + 4$
$R_y = 13$ (m)

$\vec{R} = 10\hat{i} + 13\hat{j}$ (m)

$R = \sqrt{10^2 + 13^2}$
$R = \sqrt{100 + 169}$
$R = \sqrt{269}$
$R = 16{,}40$ (m)